

Demonstrating ex-CHOCHIN: Shape/Texture-changing cylindrical interface with deformable origami tessellation

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ABSTRACT

We demonstrate ex-CHOCHIN which is a cylindrical shape/texture-changing display inspired by the chochin, a traditional Japanese foldable lantern. Ex-CHOCHIN achieves complex control over origami such as local deformation and control in the intermediate process of folding by attaching multiple mechanisms to the origami tessellation. It thereby results in flexible deformations that can be adapted to a wide range of shapes, a one-continuous surface without gaps, and even changes in texture. It creates several deformed shapes from a crease pattern, allowing flexible deformation to function as a display. We have also produced an application using ex-CHOCHIN.

During this hands-on demo at UIST, attendees will manipulate the amount of change in shape and texture and interact with ex-CHOCHIN by seeing and touching.

CCS CONCEPTS

• Human-centered computing; • Human computer interaction (CHI); • Interaction devices; • displays and imager;

KEYWORDS

Shape-changing interface, Flexible display, Tangible interaction, Origami tessellation

ACM Reference Format:

Mai Ohira, Soya Eguchi, Claire Okabe, and Hiroya Tanaka. 2022. Demonstrating ex-CHOCHIN: Shape/Texture-changing cylindrical interface with deformable origami tessellation. In *The Adjunct Publication of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22 Adjunct)*, October 29–November 02, 2022, Bend, OR, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3526114.3558626>

1 INTRODUCTION

Shape-changing displays and interfaces has the potential of enabling intuitive expression and interaction. They allow users to interact with physical 2.5D or 3D surfaces by changing their shape. In terms of haptics, shape-changing devices provide users with not

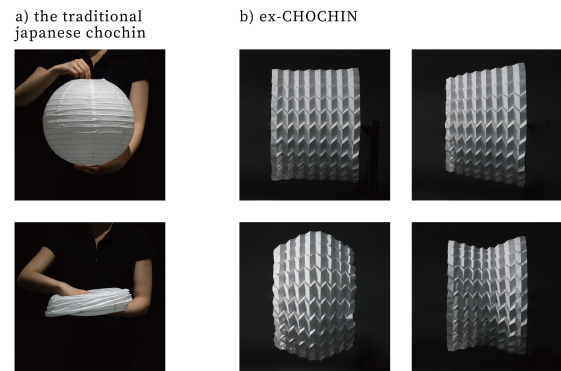


Figure 1: (a) is the traditional Japanese *chochin*, and (b) shows our ex-CHOCHIN deformed in variant shapes.

only visual but also tactile feedback, such as changes in texture, and thus function as more intuitive interfaces. There are various approaches proposed in the field of shape-changing displays. For example, inForm [1] and TRANSFORM [2] provide morphing from 2D to 2.5D by creating height differences as independently controlled pins move up and down. However, the smoothness of the undulations of their works depends on the pixel size. On the other hand, some approaches create a continuous surface by attaching membranes or origami to the surface [3-5]. These provide shape-changing displays with smoothed 2.5D undulations.

CairnFORM [6] and X-Rings [7] have developed cylindrical interfaces as 3D shape-changing devices. These displays create undulations from a cylindrical shape instead of a flat surface, thus allowing shape and texture changes in 360-degree directions. We propose ex-CHOCHIN, which generates undulation at the interface of such cylindrical shapes by using a continuous origami surface.

In inventing the ex-CHOCHIN, we were inspired by the traditional Japanese chochin: a foldable cylindrical lantern covered with a membrane. The device ex-CHOCHIN deforms the cylindrical shape into multiple three-dimensional shapes such as a truncated cone or ellipsoid, changes its texture and has a single continuous surface without gaps by using origami tessellations (Figure 1).

Various deformation approaches for origami have been proposed, including the usage of mechanical actuators [8, 9], shape-memory

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UIST '22 Adjunct, October 29–November 02, 2022, Bend, OR, USA

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ACM ISBN 978-1-4503-9321-8/22/10.

<https://doi.org/10.1145/3526114.3558626>

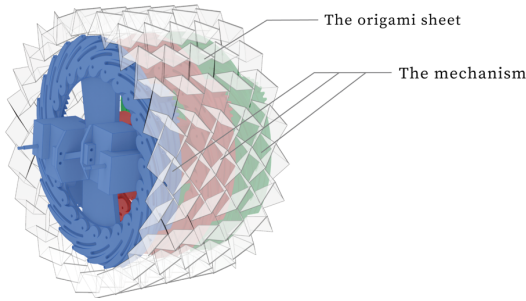


Figure 2: Basic design of ex-CHOCHIN

materials [10–12], pneumatic controls [13], and strings [14]. Mechanical actuation systems, such as the solar panel array for space satellites [8] and the transformable wheel by Hankook [9], has only two stable forms and are adapted to large objects. Shape memory materials control complex creases with simulations [10–12]. These are good for deforming to the designed shapes. However, they are not focused on holding the deformation in a specific shape or controlling the movement in the intermediate process of folding origami, such as the degree and speed of deformation. Magic-ball [13] has applied origami deformation as a pneumatic actuator. It deforms flexibly to various shapes to grasp objects. However, the deformed shapes depend on the object of grasping, not on the predefined shape. The tendon-sheath method [14] transforms the origami according to the length of the string. It has a similar concept to ex-CHOCHIN in that it transforms origami into multiple shapes.

However, while this method allows the string to be pulled and folded smaller, it does not allow for localized spreading because of the deformation when spread depends on the stiffness of the paper itself. Our concept is to transform from the same fold pattern to different shapes. Therefore, we propose an origami deformation method that uses multiple individual control mechanisms for origami tessellation.

We present the following main contributions:

- Complex control on origami, such as control in the intermediate process of folding or partial local deformation.
- Increase the flexibility of deformation shapes, such as creating multiple shapes from the same crease pattern.
- Create textures through detailed fold control.

2 FABRICATION

Figure 2 shows the configuration of ex-CHOCHIN. Multiple mechanisms are attached to the inside of the origami sheet. Each mechanism moves separately to deform the origami sheet partially and locally to create different shapes. The complexity of the possible movements depends on the number of mechanisms.

2.1 The Mechanism

This mechanism is designed to push and pull each fold from the inside, with the camera’s iris facing outward. All folds are clamped to the mechanism in order to apply an even force to the origami.

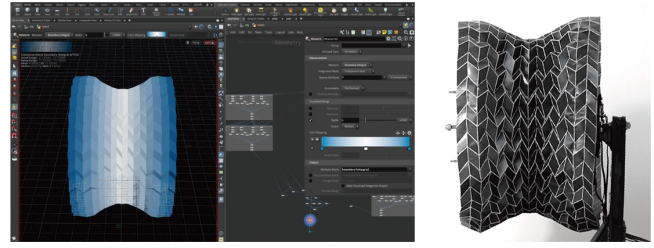


Figure 3: Design editor to define and simulate the crease patterns

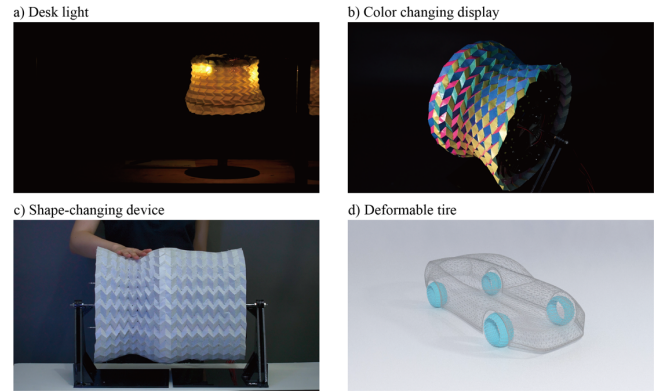


Figure 4: Applications of ex-CHOCHIN

2.2 The Crease Patterns

We adopted *Sogame-ori* [15] for the crease pattern, which is flexible and smoothly deformable. The crease pattern is created by an original software, which can predict the shape after folding. The texture is varied by changing the angle and length of the crease with parameters according to the application (Figure 3).

3 APPLICATION

The desk light in Figure 4(a) and the color changing display in Figure 4(b) are applications of visual expression using shape change. The desk light informs users of the passage of time by changing its shape and color of light. The color-changing display is a system that utilizes a visible surface that changes depending on its shape. The shape-changing interface in Figure 4(c) and the deformable tire in Figure 4(d) are texture-changing applications. Shape-changing interface is a device that provides different tactile experiences through the appearance of different undulations and textures. Deformable tire is a mock-up to demonstrate its potential to be applied to tires. It demonstrates the possibility of improving stability and ride quality by changing texture and shape in response to various road surface conditions.

4 CONCLUSION

We showed an approach for creating ex-CHOCHIN, which has new expressive capabilities and the ability to transform into a variety of shapes. Furthermore, we demonstrated applications that allow users to visually and haptically communicate information by

transforming an origami tessellation surface with a deformation mechanism. Future works include (i) adapting another mechanism to adjust the width between the mechanisms to make the orientation of the cylinders perpendicular, (ii) applying the mechanism to a swarm [16] in which many devices move in concert, and (iii) 3d-printing the mechanism consolidated and changing the size of the mechanism to make it applicable to outer space structures.

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